University of Freiburg – Mathematical Institute

Winter term 2026/2027

Comments on the Course Catalogue

Version July 16, 2026

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Study Planning

Dear Students of Mathematics,

The course catalogue provides information about the course offers of the Mathematical Institute for the current semester. Information on the course of study, which courses you can and which courses you have to pick, can be found on the information pages for each programme at https://www.math.uni-freiburg.de/lehre/. Please note that there may be different versions of examination regulations for a study programme.

You are welcome to use the advisory services of the Mathematical Institute if needed: study counseling by the programme coordinator, study counseling by the individual departments, and counseling by lecturers (office hours are listed on the personal webpages linked in the Institute's directory).

Please note:

- The two **Bachelor programmes** as well as the **Master of Education as additional subject** programmes begin with the basic lectures Analysis I and II and Linear Algebra I and II, on which most other mathematics courses are based on. In case you can only start with one of the two basic lectures due to the combination of subjects in the polyvalent dual major Bachelor programme, you can find variants for the course of study on the programme's information page.
- As an orientation requirement, the two exams for Analysis I and Linear Algebra I must be passed by the end of the 3rd semester in the **B.Sc. programme**, and at least one of the two in the **polyvalent dual major Bachelor programme**.
- There are **no further regulations regarding the structure of your individual course of study** and **no formal prerequisites for attending courses** (except for the limited number of places in each student seminar or introductory student seminar). However, you must ensure that you have the necessary prior knowledge independently.
- In the **M.Sc. Mathematics programme**, please note that you may take a maximum of two of the four oral exams with the same examiner.
- To what extent the material of advanced lectures is sufficient as **preparation for final theses and exams** or should be supplemented must be agreed upon soon enough with the supervisor of the thesis or the examiners. This applies in particular to the oral exam in the specialization module of the M.Sc. programme.

Language

Courses with the abbreviation "D" are offered in German, courses with the abbreviation "E" in English. Exercises for English lectures can often also be completed in German, and vice versa.

Presentations in student seminars can usually be given in German and English; the abbreviation "D/E" indicates this possibility.

Usability of Courses

For each course, the "Usability" section indicates in which modules from which degree programmes it can be used. For the degree programmes, the following abbreviations are used:

2HfB21	Polyvalent Dual Major Bachelor Programme
BSc21	Bachelor of Science in Mathematics, regulations of 2021
BScInfo19	Bachelor of Science in Computer Science, regulations of 2019
BScPhys22	Bachelor of Science in Physics, regulations of of 2022
MEd18	Master of Education in Mathematics
MEdual24	Master of Education – "(Gymnasium) with integrated practice"
MEH21	Master of Education, Mathematics as an additional subject with 120 ECTS points
MEB21	Master of Education, Mathematics as an additional subject with 90 ECTS points
MSc14	Master of Science in Mathematics
MScData24	Master of Science in Mathematics in Data and Technology

As a general rule, no courses that were already used in the underlying Bachelor programme may be completed in a Master programme. If you have any questions, please contact the programme coordination.

Please note:

- It is allowed to use higher-level lectures, typically offered for the M.Sc. Mathematics programme, for electives in other study programmes; however, due to the required prior knowledge, they will only be suitable in exceptional cases. If a course can be used for a module, does not necessarily mean that it is suitable for a module. Conversely, extreme cases are not listed though possible (e.g. a lecture such as "Differential Geometry II" as a specialisation module in the M.Ed. study programme).
- In the B.Sc. Mathematics, in addition to the compulsory area at least three 4-hour lectures with 2-hour exercises (9 ECTS points each) must be completed. At least one of these must be from the field of pure mathematics. You can see which of the lectures count as pure mathematics by checking whether it is approved for the module "Pure Mathematics" in the M.Sc. Mathematics programme.

Examination Requirements and Pass/Fail Assessments

The section "Usability" will be supplemented at the beginning of the lecture period with information on which graded examinations and pass/fail assessments are required for its use in the respective module or study area. This information supplements the module handbooks in terms of examination and accreditation law and will be approved by the Mathematics Study Commission.

Please note:

- Deviations from the specified type of graded examination are permitted if, due to circumstances beyond the examiner's control, the intended type of examination is not suitable or would involve disproportionate effort. The same applies to pass/fail assessments.
- If a course is approved as an elective module in a non-listed study programme, the requirements follow
  - the elective module of the B.Sc. programme, if examination achievements are required,
  - the elective module of the M.Sc. programme, if only pass/fail assessments are required.

  If the corresponding modules are not offered, please contact the programme coordination of the Mathematical Institute.
- If written exercise assignments are required as a study achievement, these are usually weekly exercise assignments, or bi-weekly in the case of a one-hour exercise. Depending on the start, end, rhythm, and individual breaks, there can be between 5 and 14 exercise sheets. The number of points achievable per exercise sheet can vary.
- For programming exercises, this applies analogously to programming assignments.

Research Areas for Final Theses

Information on Bachelor and Master theses in Mathematics can be found here:

https://www.math.uni-freiburg.de/lehre/en/page/abschlussarbeiten/

The following list gives you an overview of the fields from which the professors and lecturers of the Mathematical Institute typically assign topics for final theses. If you are interested in a thesis, please arrange an appointment early!

Prof. Dr. **Sören Bartels**	Applied Mathematics, Partial Differential Equations, and Numerics
Prof. Dr. **Harald Binder**	Medical Biometrics and Applied Statistics
JProf. Dr. **David Criens**	Stochastic Analysis, Probability Theory, and Financial Mathematics
Prof. Dr. **Moritz Diehl**	Numerics, Optimization, Optimal Control
Prof. Dr. **Patrick W. Dondl**	Applied Mathematics, Calculus of Variations, Partial Differential Equations, and Numerics
PD Dr. **Giuseppe Genovese**	Probability Theory, Machine Learning, Statistical Mechanics
Prof. Dr. **Sebastian Goette**	Differential Geometry, Topology, and Global Analysis
Prof. Dr. **Nadine Große**	Differential Geometry and Global Analysis
Prof. Dr. **Annette Huber-Klawitter**	Algebraic Geometry and Number Theory
PD Dr. **Markus Junker**	Mathematical Logic, Model Theory
Prof. Dr. **Stefan Kebekus**	Algebra, Complex Analysis, Complex and Algebraic Geometry
Prof. Dr. **Ernst Kuwert**	Partial Differential Equations, Calculus of Variations
Prof. Dr. **Eva Lütkebohmert-Holtz**	Financial Mathematics, Risk Management and Regulation
Prof. Dr. **Amador Mart’ın Pizarro**	Mathematical Logic, especially Model Theory
Prof. Dr. **Heike Mildenberger**	Mathematical Logic, especially Set Theory and Infinite Combinatorics
JProf. Dr. **Abhishek Oswal**	Algebra, Algebraic Geometry, and Number Theory
Prof. Dr. **Peter Pfaffelhuber**	Stochastics, Biomathematics
Prof. Dr. **Angelika Rohde**	Mathematical Statistics, Probability Theory
Prof. Dr. **Michael Růžička**	Applied Mathematics and Partial Differential Equations
Prof. Dr. **Chiara Saffirio**	Mathematical Physics
JProf. Dr. **Diyora Salimova**	Applied Mathematics, Partial Differential Equations, Machine Learning, and Numerics
Prof. Dr. **Thorsten Schmidt**	Probability Theory, Financial and Actuarial Mathematics, Machine Learning
Prof. Dr. **Wolfgang Soergel**	Algebra and Representation Theory
Prof. Dr. **Guofang Wang**	Partial Differential Equations, Calculus of Variations

Offers from EUCOR Partner Universities

As part of the EUCOR cooperation, you can attend courses at the partner universities *University of Basel*, *Karlsruhe Institute of Technology*, *Université Haute-Alsace* in Mulhouse, and the *Université de Strasbourg*. The procedure is explained in detail on this information page.

In particular, Basel and Strasbourg offer interesting additions to our lecture programme at the master's level. Credits can be recognized within the framework of the respective examination regulations, especially in the elective (required) area of the B.Sc. and M.Sc. programmes. Please discuss possible credits in advance with the programme coordination!

The costs for travel by train, bus, and tram can be subsidized by EUCOR.

Basel

Institute: The Department of Mathematics and Computer Science at the University of Basel offers eight research groups in mathematics: Algebraic Geometry, Number Theory, Analysis, Numerics, Computational Mathematics, Probability Theory, Mathematical Physics, and Statistical Science.

Lecture Offerings: The pages with the lecture offerings for the Bachelor and the lecture offerings for the Master seem to correspond most closely to our mathematics lecture directory. The general lecture directory of the university can be found here: `https://vorlesungsverzeichnis.unibas.ch/de/semester-planung`

Dates: In Basel, the autumn semester begins in mid-September and ends at the end of December, the spring semester runs from mid-February to the end of May.

Travel: The University of Basel is best reached by train: The train ride to the Badischer Bahnhof takes about 45-60 minutes by local transport, 30 minutes by ICE. Then take tram 6 towards *Allschwil Dorf* to the *Schifflände* stop (about 10 minutes).

Strasbourg

Institute: In Strasbourg, there is a large *Institut de recherche mathématique avancée* (IRMA), which is divided into seven *équipes*: Analyse; Arithmétique et géométrie algébrique; Algèbre, représentations, topologie; Géométrie; Modélisation et contrôle; Probabilités und Statistique. Seminars and working groups (*groupes de travail*) are announced on the institute's website.

Lecture Offerings: Participation of Freiburg students in the offers of the second year of the Master M2 is highly welcome. Depending on prior knowledge, the lectures are suitable for our students from the 3rd year of study onwards. The lecture language is a priori French, but a switch to English is gladly possible upon request, please arrange in advance. In Strasbourg, a different focus topic is offered in the M2 annually, in 2026/27 it is: *Probabilités*.

General lecture directories typically do not exist in France.

Dates: In France, the *1st semester* runs from early September to late December and the *2nd semester* from late January to mid-May. A more precise schedule will only be available in September. The timetables are flexible; they can usually accommodate the needs of Freiburg students.

Travel: The *Université de Strasbourg* is best reached by car (about one hour). Alternatively, there is a very affordable connection with Flixbus to the *Place de l'Étoile*. The train ride to the main station in Strasbourg takes about 1h40 by local transport, 1h10 by ICE. Then take tram line C towards *Neuhof, Rodolphe Reuss* to the *Universités* stop.

For further information and organizational assistance, please contact:

in Freiburg: Prof. Dr. Annette Huber-Klawitter annette.huber@math.uni-freiburg.de

in Strasbourg: Prof. Carlo Gasbarri, Coordinator of the M2 gasbarri@math.unistra.fr
or the respective course coordinators.

1a. Compulsory Lectures of the various Study Programmes

Analysis I

Guofang Wang, Assistant: Xuwen Zhang in German
Lecture: Tue, Wed, 8–10 h, HS Rundbau, Albertstr. 21
Tutorial: 2 hours, various dates
Sit-in exam: date to be announced

Content:

Analysis I is one of the two basic lectures in the mathematics course. It deals with concepts based on the notion of limit. The central topics are: induction, real and complex numbers, convergence of sequences and series, completeness, exponential function and trigonometric functions, continuity, derivation of functions of one variable and regulated integrals.

Literature:

To be announced in the lecture.

Prerequisites:

Required: High school mathematics.
Attendance of the pre-course (for students in mathematics) is recommended.

Usable in the following modules:

Analysis (2HfB21, BSc21, MEH21, MEB21)
Analysis I (BScInfo, BScPhys)

Linear Algebra I

Wolfgang Soergel, Assistant: Damian Sercombe in German
Lecture: Mon, Thu, 8–10 h, HS Rundbau, Albertstr. 21
Tutorial: 2 hours, various dates
Sit-in exam: date to be announced

Content:

Linear Algebra I is one of the two introductory lectures in the mathematics degree programme that form the basis for further courses. Topics covered include: fundamental concepts (in particular fundamental concepts of set theory and equivalence relations), groups, fields, vector spaces over arbitrary fields, basis and dimension, linear mappings and transformation matrix, matrix calculus, linear systems of equations, Gaussian elimination, linear forms, dual space, quotient vector spaces and homomorphism theorem, determinant, eigenvalues, polynomials, characteristic polynomial, diagonalizability, affine spaces. The background to the mathematical content is explained in terms of ideas and the history of mathematics.

Literature:

To be announced in the lecture.

Prerequisites:

Required: High school mathematics.
Attendance of the pre-course (for students in mathematics) is recommended.

Usable in the following modules:

Linear Algebra (2HfB21, BSc21, MEH21)
Linear Algebra (MEB21)
Linear Algebra I (BScInfo, BScPhys)

Numerics I

Sören Bartels, Assistant: Jonathan Brugger in German
Lecture: Wed, 14–16 h, HS Weismann-Haus, Albertstr. 21a
Tutorial: 2 hours fortnightly, various dates

Content:

Numerics is a sub-discipline of mathematics that deals with the practical solution of mathematical problems. In most cases, problems are not solved exactly but approximately, for which a sensible compromise between accuracy and computational effort must be found. The first part of the two-semester course focuses on questions of linear algebra such as solving linear systems of equations and determining the eigenvalues of a matrix.

Literature:

- S. Bartels: *Numerik 3x9*. Springer, 2016.
- R. Plato: *Numerische Mathematik kompakt*. Vieweg, 2006.
- R. Schaback, H. Wendland: *Numerische Mathematik*. Springer, 2004.
- J. Stoer, R. Burlisch: *Numerische Mathematik I, II*. Springer, 2007, 2005.
- G. Hämmerlin, K.-H. Hoffmann: *Numerische Mathematik*. Springer, 1990.
- P. Deuflhard, A. Hohmann, F. Bornemann: *Numerische Mathematik I, II*. DeGruyter, 2003.

Prerequisites:

Required: Linear Algebra I
Recommended: Linear Algebra II and Analysis I (required for Numerics II)

Remarks:

Programming exercises (*Praktische Übung*) are offered to accompany the lecture. They are compulsory in the B.Sc. study programme in Mathematics and recommended for the other study programmes. They take place every 14 days, alternating with the lecture's tutorial.

Usable in the following modules:

Numerics (BSc21)
Numerics (2HfB21, MEH21)
Numerics I (MEB21)

Elementary Probability Theory I

Ernst August v. Hammerstein, Assistant: Sebastian Stroppel in German
Lecture: Tue, 14–16 h, HS Weismann-Haus, Albertstr. 21a
Tutorial: 2 hours fortnightly, various dates
Sit-in exam 23.02., ,

Content:

Stochastic is, to put it loosely, the "mathematics of chance", about which – possibly contrary to first impressions – many precise and not at all random statements can be formulated and proven. The aim of the lecture is to give an introduction to stochastic modeling, to explain some basic concepts and results of Stochastic and to illustrate them with examples. It is also intended as a motivating preparation for the lecture "Probability Theory" in the summer semester, especially for students in the B.Sc. in Mathematics. Topics covered include: Discrete and continuous random variables, probability spaces and measures, combinatorics, expected value, variance, correlation, generating functions, conditional probability, independence, weak law of large numbers, central limit theorem.

The lecture Elementary Probability Theory II in the summer term will mainly be devoted to statistical topics.

Literature:

- L. Dümbgen: *Stochastik für Informatiker*, Springer, 2003.
- H.-O. Georgii: *Stochastik: Einführung in die Wahrscheinlichkeitstheorie und Statistik* (5. Auflage), De Gruyter, 2015.
- N. Henze: *Stochastik für Einsteiger* (14. Auflage), Springer Spektrum, 2023.
- N. Henze: *Stochastik: Eine Einführung mit Grundzügen der Maßtheorie* (2. Auflage), Springer Spektrum, 2024.
- G. Kersting, A. Wakolbinger: *Elementare Stochastik* (2. Auflage), Birkhäuser, 2010.

Prerequisites:

Required: Linear Algebra I, Analysis I and II.
Note that Linear Algebra I can be attended in parallel.

Remarks:

If you are interested in a practical, computer-supported implementation of individual lecture contents, participation in the regularly offered programming exercise "Python for Data Analysis" is also recommended (in parallel or subsequently).

Usable in the following modules:

Elementary Probabilty Theory (2HfB21, MEH21)
Elementary Probability Theory I (BSc21, MEB21, MEdual24)

Basics in Applied Mathematics

Peter Pfaffelhuber, Diyora Salimova in English
Lecture: Tue, Thu, 10–12 h, HS II, Albertstr. 23b
Tutorial: 2 hours, date to be determined and announced in class
Programming exercise: 2 hours, date to be determined

Content:

This course provides an introduction into the basic concepts, notions, definitions and results in probability theory, numerics and optimization, accompanied with programming projects in Python. Besides deepen mathematical skills in principle, the course lays the foundation of further classes in these three areas.

Literature:

Lecture notes will be provided.

Prerequisites:

None that go beyond admission to the degree programme.

Usable in the following modules:

Basics in Applied Mathematics (MScData24)

Further Chapters in Analysis

Susanne Knies, Assistant: Jonah Reuß in German
Lecture: Thu, 14–16 h, HS Weismann-Haus, Albertstr. 21a
Tutorial: 2 hours, various dates
Sit-in exam 22.02., 10:00–12:00, ,

Content:

This compulsory lecture for students of teacher education in the M.Ed. builds on the basic lectures Analysis I and II and supplements them with the following two main topics:

Multidimensional integration: The one-dimensional Riemann integral known from Analysis I is generalized to real-valued functions of several variables, for which a suitable instrument for measuring the content/volume of multidimensional sets is first introduced with the Jordan content. Then the classical integral theorems (transformation theorem, Fubini's theorem) are derived, and path and surface integrals are considered. With the help of the divergence and rotation of vector fields, the two aforementioned integral types can be related to each other using the integral theorems of Gauß and Stokes, which considerably simplifies the calculations in practical applications.

Multiple integration: Jordan content in $\mathbb{R}^n$, Fubini's theorem, transformation theorem, divergence and rotation of vector fields, path and surface integrals in $\mathbb{R}^3$, Gauss' theorem, Stokes' theorem.

Complex analysis: Introduction to the theory of holomorphic functions, Cauchy's integral theorem, Cauchy's integral formula and applications.

Literature:

- K. Königsberger: *Analysis 2*, 5. Auflage., Springer, 2004.
- W. Walter: *Analysis 2*, 5. Auflage, Springer, 2002.
- E. Freitag, R. Busam: *Funktionentheorie I*, 4. Auflage, Springer, 2006.
- R. Remmert, G. Schumacher: *Funktionentheorie 1*. 5. Auflage, Springer, 2002.

Prerequisites:

Required: Analysis I and II, Linear Algebra I and II

Usable in the following modules:

Further Chapters in Analysis (MEd18, MEH21, MEdual24)

Aspects of Analysis and Linear Algebra in Secondary School Mathematics

Katharina Böcherer-Linder, Markus Junker in German
Wed, 13–15 h, SR 404, Ernst-Zermelo-Str. 1

Content:

This newly designed course will address topics from lectures on Analysis and Linear Algebra (such as limits, continuity, geometric mappings) and examine how these topics are covered in secondary school and to what extent a university-level understanding of mathematics helps in understanding secondary school mathematics.

The event is planned as an interactive student seminar in which participants prepare case studies that are then discussed together. Assessment will be based on regular attendance and the presentations and elaborations of the case studies.

Prerequisites:

Introductory lectures in Analysis and Linear Algebra

Usable in the following modules:

Supplementary Module in Mathematics (MEd18)
High-school Oriented Aspects of Analysis and Linear Algebra (MEdual24)

1b. Advanced 4-hour Lectures

Algebraic Number Theory

Abhishek Oswal, Assistant: Ben Snodgrass in English
Lecture: Mon, Wed, 14–16 h, HS II, Albertstr. 23b
Tutorial: 2 hours, date to be determined and announced in class

Content:

Short description of topics: Number fields, Prime decomposition in Dedekind domains, Ideal class groups, Unit groups, Dirichlet's unit theorem, local fields, valuations, decomposition and inertia groups, introduction to class field theory.

Literature:

Jürgen Neukirch: *Algebraic Number Theory*, Springer, 1999.

Prerequisites:

Required: Algebra and Number Theory

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Mathematical Specialisation (MEd18, MEH21)
Pure Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective (MScData24)

Algebra and Number Theory

Stefan Kebekus, Assistant: Riccardo Tosi in German
Lecture: Tue, Thu, 10–12 h, HS Weismann-Haus, Albertstr. 21a
Tutorial: 2 hours, various dates
Sit-in exam: date to be announced

Content:

This lecture continues the Linear Algebra courses. It treats groups, rings, fields and applications in the number theory and geometry. The highlights of the lecture are the classification of finite fields, the impossibility of the trisection of angles with circle and ruler, the non-existence of a solution formula for the general equations of fifth degree and the quadratic reciprocity law.

Literature:

- Michael Artin: *Algebra*, Birkhäuser 1998.
- Siegfried Bosch: *Algebra* (8th edition), Springer Spektrum 2013.
- Serge Lang: *Algebra* (3rd edition), Springer 2002.
- Wolfgang Soergel: lecture notes *Algebra und Zahlentheorie*

Prerequisites:

Linear Algebra I and II

Usable in the following modules:

Algebra and Number Theory (2HfB21, MEH21)
Compulsory Elective in Mathematics (BSc21)
Introduction to Algebra and Number Theory (MEB21)
Algebra and Number Theory (MEdual24)
Pure Mathematics (MSc14)
Elective (MSc14)
Elective (MScData24)

Analysis III

Ernst Kuwert, Assistant: Mingwei Zhang in German
Lecture: Mon, Wed, 10–12 h, HS Weismann-Haus, Albertstr. 21a
Tutorial: 2 hours, various dates
Sit-in exam: date to be announced

Content:

This lecture provides an introduction to Lebesgue's theory of multidimensional measures and integration. The focus is on general measures and integrals, convergence theorems, integration in $\mathbb{R}^n$, the transformation theorem and Gauss's theorem, and possibly also differential forms.

The lecture is relevant for further study in analysis, applied mathematics, stochastics, probability theory and geometry, as well as in physics.

Prerequisites:

Analysis I and II, Linear Algebra I and II

Usable in the following modules:

Elective (Option Area) (2HfB21)
Analysis III (BSc21)
Mathematical Specialisation (MEd18, MEH21)
Elective in Data (MScData24)

Differential Geometry

Nadine Große, Assistant: Jonah Reuß — in English
Lecture: Tue, Thu, 8–10 h, HS II, Albertstr. 23b
Tutorial: 2 hours, date to be determined and announced in class

Content:

Differential geometry investigates geometric properties of curved spaces using methods of differential calculus. It has applications in other areas of mathematics and in physics, such as theoretical mechanics and the theory of relativity.

This lecture provides an introduction to (semi-)Riemannian geometry. In particular, it will focus on geodesics and the Riemannian curvature tensor.

Literature:

- Barrett O'Neill: *Semi-Riemannian Geometry with Applications to Relativity*, Academic Press, 1983.
- J.M. Lee: *Introduction to Smooth Manifolds*, Springer, 2003.
- M.P. do Carmo: *Riemannian Geometry*, Birkhäuser, 1992

Prerequisites:

Required: Analysis I–III, Lineare Algebra I and II
Prior exposure to curves and surfaces ("Kurven und Flächen") and topology is beneficial.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Mathematical Specialisation (MEd18, MEH21)
Pure Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective (MScData24)

Complex Analysis

Ernst August v. Hammerstein — in German

Lecture: Mon, Wed, 8–10 h, HS Weismann-Haus, Albertstr. 21a
Tutorial: 2 hours, date to be determined and announced in class
Sit-in exam: date to be announced

Content:

Complex Analysis is a classical branch of advanced mathematics that was largely established during the 19th century through the pioneering work of Cauchy, Riemann, and Weierstraß. It extends the concepts of differentiability and integrability, familiar from analysis, to complex-valued functions $f : \mathbb{C} \to \mathbb{C}$ and examines the resulting consequences, which are far richer than one might initially assume and lead to a theory that is often more elegant and sometimes even simpler. This theory is fundamental to many advanced areas of mathematics, particularly number theory and algebraic geometry, and its applications extend to probability theory, functional analysis, and mathematical physics.

The starting point for all these considerations is that complex numbers can be identified with points in $\mathbb{R}^2$ (the complex or Gaussian plane), and thus complex differentiable functions can be identified with real-valued functions $g : \mathbb{R}^2 \to \mathbb{R}^2$ that satisfy the so-called Cauchy–Riemann differential equations. The surprising results of complex analysis can ultimately be traced back to the particularly elegant properties of these differential equations. Some of these properties state, for example, that complex-differentiable functions are continuously differentiable infinitely often, they can be represented locally as power series, and their values within a circular disc are already determined by those on its boundary.

Central topics of the lecture will include, among others, the Cauchy–Riemann differential equations, Cauchy's integral theorem, Cauchy's integral formula, the maximum principle, Laurent series, the residue theorem, and Riemann's mapping theorem. Depending on students' interest and remaining time, more advanced topics such as Picard's theorems, doubly periodic functions, or applications in number theory may also be covered.

Literature:

- F. Bornemann: *Funktionentheorie* (2. Auflage), Birkhäuser, 2016.
- W. Fischer, I. Lieb: *Funktionentheorie* (9. Auflage), Springer Vieweg, 2005.
- E. Freitag, R. Busam: *Funktionentheorie 1* (4. Auflage), Springer, 2006.
- K. Jänich: *Funktionentheorie. Eine Einführung* (6. Auflage), Springer, 2004.
- R. Remmert, G. Schumacher: *Funktionentheorie 1* (5. Auflage), Springer, 2002.
- D.A. Salamon: *Funktionentheorie*, Birkhäuser, 2012.

Prerequisites:

Analysis I and II, Linear Algebra I

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Mathematical Specialisation (MEd18, MEH21)
Pure Mathematics (MSc14)
Elective (MSc14)
Elective (MScData24)

Introduction to Theory and Numerics of Partial Differential Equations

Michael Růžička, Assistant: Alen Kushova in English
Lecture: Mon, Wed, 10–12 h, HS II, Albertstr. 23b
Tutorial: 2 hours, date to be determined and announced in class

Content:

This lecture is the first in a series of sequential lectures on the theory and numeric of partial differential equations.

Partial differential equations often serve as models for physical processes, e.g., in determining a temperature distribution, in describing vibrations of membranes, or fluid flows.

In this lecture, we will focus on elliptic differential equations. We will cover both the classical existence theory and the modern theory of solvability for such equations. Even if one has explicit solution formulas for simple problems, these can rarely be computed in practice. Therefore, it is important to compute numerically approximate solutions and to verify that they converge appropriately toward the exact solution. To this end, the lecture will present the corresponding theory of finite elements.

Programming exercises will be offered in parallel with the lecture (see the comments on the programming exercises).

Literature:

- D. Braess: *Finite Elemente* (5. Auflage), Springer Spektrum, 2013.
- G. Dziuk: *Theorie und Numerik partieller Differentialgleichungen*, De Gruyter, 2010.
- L. C. Evans: *Partial Differential Equations* (Second Edition), Graduate Studies in Mathematics 19, AMS, 2010.

Prerequisites:

Required: Analysis I to III, Linear Algebra I and II
Recommended: Functional Analysis

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Mathematical Specialisation (MEd18, MEH21)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Advanced Lecture in Numerics (MScData24)
Elective in Data (MScData24)

Mathematical Statistics

in English

Lecture: Wed, Thu, 10–12 h, SR 226, Hermann-Herder-Str. 10
Tutorial: 2 hours, date to be determined and announced in class

Content:

The lecture builds on basic knowledge about Probability Theory. The fundamental problem of statistics is to infer from a sample of observations as precise as possible statements about the data-generating process or the underlying distributions of the data. For this purpose, the most important methods from statistical decision theory such as test and estimation methods are introduced in the lecture.

Key words hereto include Bayes estimators and tests, Neyman-Pearson test theory, maximum likelihood estimators, UMVU estimators, exponential families, linear models. Other topics include ordering principles for reducing the complexity of models (sufficiency and invariance).

Statistical methods and procedures are used not only in the natural sciences and medicine, but in almost all areas in which data is collected and analyzed. This includes, for example, economics ("econometrics") and the social sciences (especially psychology). However, in the context of this lecture, we will focus less on applications, but – as the name suggests – more on the mathematical justification of the methods.

Literature:

- C. Czado, T. Schmidt: *Mathematische Statistik*, Springer, 2011.
- E.L. Lehmann, J.P. Romano: *Testing Statistical Hypotheses (Fourth Edition)*, Springer, 2022.
- E.L. Lehmann, G. Casella: *Theory of Point Estimation, Second Edition*, Springer, 1998.
- L. Rüschendorf: *Mathematische Statistik*, Springer Spektrum, 2014.
- M. J. Schervish: *Theory of Statistics*, Springer, 1995.
- J. Shao: *Mathematical Statistics*, Springer, 2003.
- H. Witting: *Mathematische Statistik I*, Teubner, 1985.

Prerequisites:

Probability Theory (in particular measure theory and conditional probabilities/expectations)

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Advanced Lecture in Stochastics (MScData24)
Elective in Data (MScData24)

Probabilistic Machine Learning

Giuseppe Genovese, Assistant: Roger Bader in English
Lecture: Tue, Fri, 12–14 h, SR 127, Ernst-Zermelo-Str. 1
Tutorial: 2 hours, various dates
Programming exercise: 2 hours, date to be determined

Content:

The goal of the course is to provide a mathematical treatment of deep neural networks and energy models, that are the building blocks of many modern machine learning architectures. About neural networks we will study the basics of statistical learning theory, the back-propagation algorithm and stochastic gradient descent, the benefits of depth. About energy models we will cover some of the most used learning and sampling algorithms. In the exercise classes, besides solving theoretical problems, there will be some Python programming sessions to implement the models introduced in the lectures.

Prerequisites:

Probability Theory I
Basic knowledge of Markov chains is useful for some part of the course.

Remarks:

References will be provided during the course.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Advanced Lecture in Stochastics (MScData24)
Elective in Data (MScData24)

Probability Theory II – Stochastic Processes

David Criens in English
Lecture: Tue, Thu, 12–14 h, HS II, Albertstr. 23b
Tutorial: 2 hours, date to be determined and announced in class

Content:

A stochastic process $(X_t)_{t\in T}$ is a family of random variables, where mostly the situation $T = \mathbb{N}$ or $T = [0, 1]$ is studied. Basic examples include stationary time series, the Poisson process and Brownian motion as well as processes derived from those. The lecture includes ergodic theory and its applications, Brownian motion and especially the study of its path properties, the elegant concept of weak convergence on Polish spaces as well as functional limit theorems. Finally, we introduce stochastic integration with respect to local martingales, based on the continuous time version of the martingale transform.

Literature:

- Achim Klenke: *Probability Theory – a comprehensive course*
- Olav Kallenberg: *Foundations of modern probability*

Prerequisites:

Probability Theory I

Remarks:

In the summer semester, this course will be continued with the lecture Stochastic Analysis.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Advanced Lecture in Stochastics (MScData24)
Elective in Data (MScData24)

Probability Theory IV: Stochastic Partial Differential Equations

Angelika Rohde, Assistant: Johannes Brutsche in English
Lecture: Mon, Wed, 12–14 h, HS II, Albertstr. 23b
Tutorial: 2 hours, date to be determined and announced in class

Content:

This lecture provides a systematic and self-contained introduction to the fascinating and elegant theory of stochastic evolution equations in infinite-dimensional spaces. This includes, in particular, a presentation of the essential properties of probability measures on separable Banach and Hilbert spaces, as well as the corresponding integration theory for defining the expectation of Banach-space-valued random variables. As a second central part, we treat existence and uniqueness results for stochastic partial differential equations and investigate qualitative properties of their solutions. Well-known examples of such equations include, for instance, the stochastic wave equation and the stochastic Schrödinger equation.

Literature:

- da Prato und Zabczyk: *Stochastic Equations in Infinite Dimensions*, 2nd edition, Cambridge University Press 2014
- Liu und Röckner: *Stochastic Partial Differential Equations: An Introduction*, Universitext, Springer, 2015

Prerequisites:

required: Probability Theory II
helpful: Probability Theory III

Remarks:

Master's thesis topics can be assigned on the basis of this lecture. This includes, in particular, topics on the statistical theory of SPDEs, which is still a young but rapidly growing field of research.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Advanced Lecture in Stochastics (MScData24)
Elective in Data (MScData24)

Set Theory: Independence Proofs

Heike Mildenberger in English
Lecture: Mon, Wed, 10–12 h, SR 404, Ernst-Zermelo-Str. 1
Tutorial: 2 hours, date to be determined and announced in class

Content:

How does one show that something cannot be proved? More precisely: how does one show that a statement does not follow from certain axioms? The lecture begins with a brief introduction to the most common axiomatic systems in mathematics: the Zermelo–Fraenkel system with the Axiom of Choice (ZFC) and the axiomatic system of von Neumann, Bernays and Gödel (NBG). The axioms shape our understanding of the possible definable – or perhaps less constructively given – mathematical objects. However, they do not paint a complete picture of a single mathematical universe. The list of derivable mathematical statements is incomplete: for some φ , neither φ nor its negation can be proven from ZFC. We say that 'φ is independent of ZFC'. The best-known statement independent of ZFC is the Continuum Hypothesis, which states that there are exactly $\aleph_1$ real numbers.

This lecture introduces the technique of independence proofs. After starting with simple forcing techniques for cardinal number exponentiation, we will explore ZF-models without the Axiom of Choice and iterated forcing (e.g. to prove the relative consistency of Martin's Axiom). Lecture notes are available.

Literature:

- H.-D. Ebbinghaus, *Einführung in die Mengenlehre.* 4. Auflage, 2003.
- Paul Eklof, Alan Mekler, *Almost Free Modules, Revised Edition*, North-Holland, 2002.
- Lorenz Halbeisen, *Combinatorial Set Theory. With a Gentle Introduction to Forcing*, Springer, 2012.
- Thomas Jech, *Set Theory. The Third Millenium* Edition, Springer, 2001.
- Kenneth Kunen, *Set Theory, An Introduction to Independence Proofs*, North-Holland, 1980.
- Kenneth Kunen, *Set Theory.* Second Edition, College Publications, 2013.
- Saharon Shelah, *Proper and Improper Forcing*, Springer, 1998.

Prerequisites:

Prerequisites: Introductory lectures
Useful background knowledge: Mathematical logic

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Pure Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective (MScData24)

Valued Fields

Amador Martín Pizarro in English
Lecture: Tue, Thu, 12–14 h, SR 404, Ernst-Zermelo-Str. 1
Tutorial: 2 hours, date to be determined and announced in class

Content:

We obtain the field $\mathbb{R}$ of real numbers as the completion of $\mathbb{Q}$ with respect to the standard absolute value, by adding the limit of every Cauchy sequence to that sequence. For a prime number p, we define the p-adic absolute value of a non-zero rational number q as

$$|q|_p = e^{-\operatorname{ord}_p(q)}$$

where $\operatorname{ord}_p(q) = n$ if $q = p^n \cdot \frac{a}{b}$ such that p divides neither a nor b. The p-adic absolute value satisfies a stronger form of the triangle inequality, and every integer has a p-adic absolute value of at most 1. The completion of $\mathbb{Q}$ with respect to $||_p$ is the field $\mathbb{Q}_p$ of p-adic numbers. Thus, amongst other things, we obtain an element in $\mathbb{Q}_p$ as the limit of the partial series

$$s_n = \sum_{k \leqslant n} p^k.$$

In this lecture, we shall investigate properties of the p-adic absolute value and its valuation ord_p. The aim of the lecture is to give a (nearly) positive answer to a conjecture by Emil Artin: Artin claimed that every non-trivial polynomial over $\mathbb{Q}_p$ of degree d in more than $d^2 + 1$ variables has a non-trivial zero.

Literature:

P. L. Clark:

Local Fields.

A. J. Engler, A. Prestel: Valued Fields, Springer, 2005.

F. V. Kuhlmann: *Valuation Theory*.

Prerequisites:

Course 'Algebra and Number Theory'

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Pure Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective (MScData24)

Reading courses

All professors and 'Privatdozenten' of the Mathematical Institute Talk/participation possible in German and English, in English
Dates by arrangement

Content:

In a reading course, the material of a four-hour lecture is studied in supervised self-study. In rare cases, this may take place as part of a course; however, reading courses are not usually listed in the course catalogue. If you are interested, please contact a professor or a private lecturer before the start of the course; typically, this will be the supervisor of your Master's thesis, as the reading course ideally serves as preparation for the Master's thesis (both in the M.Sc. and the M.Ed. programmes).

The content of the reading course, the specific details, and the coursework requirements will be determined by the supervisor at the beginning of the lecture period. The workload should be equivalent to that of a four-hour lecture with exercises.

Usable in the following modules:

Reading Course (MEd18, MEH21)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)

1c. Advanced 2-hour Lectures

Advanced Course in Schemes

Damian Sercombe in English
Lecture: Tue, 10–12 h, SR 127, Ernst-Zermelo-Str. 1

Content:

Schemes represent the generalization of varieties to arbitrary base rings. Master's and doctoral students specializing in algebraic or arithmetic geometry cannot avoid this theory. Traditionally, students master the material through independent study; this course aims to support that process. We will rely on Hartshorne's established text (Chapter II and parts of Chapter III), covering topics such as sheaves, schemes, separated and proper morphisms, projective morphisms, differentials, flat and smooth morphisms, line bundles and divisors, and sheaf cohomology.

The lectures will present the key aspects of each topic, while students are expected to work through the details via independent study of the literature. One tutorial session will provide an opportunity to discuss the reading material, while a second session will be dedicated to answering questions and discussing exercises. The scope and workload will correspond to a four-hour lecture course. The course will be conducted in English.

Literature:

- R. Hartshorne: *Algebraic Geometry*
- Görtz and Wedhorn: *Algebraic Geometry I – Schemes*

Prerequisites:

Commutative Algebra

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Pure Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective (MScData24)

Futures and Options

Eva Lütkebohmert-Holtz in English
Lecture: Mon, 10–12 h, -, -
Exercise session: Tue, 8–10 h, -, -
Sit-in exam: date to be announced

Content:

This course covers an introduction to financial markets and products. Besides futures and standard put and call options of European and American type we also discuss interest-rate sensitive instruments such as swaps.

For the valuation of financial derivatives we first introduce financial models in discrete time as the Cox–Ross–Rubinstein model and explain basic principles of risk-neutral valuation. Finally, we will discuss the famous Black–Scholes model which represents a continuous time model for option pricing.

Literature:

- D. M. Chance, R. Brooks: *An Introduction to Derivatives and Risk Management* (10th edition), Cengage, 2016.
- J. C. Hull: *Options, Futures, and other Derivatives* (11th global edition), Pearson, 2021.
- S. E. Shreve: *Stochastic Calculus for Finance I: The Binomial Asset Pricing Model*, Springer, 2004.
- R. A. Strong: *Derivatives. An Introduction* (Second edition), South-Western, 2004.

Prerequisites:

Elementary Probability Theory I

Remarks:

The course is offered for the first year in the *Finance profile* of the M.Sc. Economics programme as well as for students of M.Sc. and B.Sc. Mathematics, M.Sc. Mathematics in Data and Technology and M.Sc. Volkswirtschaftslehre. In the M.Sc. Mathematics, it can also count as elective in economics for the specialization in financial mathematics. For students who are currently in the B.Sc. Mathematics programme, but plan to continue with this special profile, it is therefore recommended to credit this course for the latter profile and not for B.Sc. Mathematics.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective in Data (MScData24)

Harmonic Analysis

Chiara Saffirio, Assistant: Phillip Pflaum in English
Lecture: Mon, 12–14 h, SR 404, Ernst-Zermelo-Str. 1
Tutorial: 2 hours, date to be determined and announced in class

Content:

Topics on Fourier series (convergence in norm; multipliers and almost everywhere convergence; applications to geometry and PDEs, applications to number theory and ergodic theory)

Literature:

L. Grafakos *Classical Fourier Analysis*, https://doi.org/10.1007/978-1-4939-1194-3.

Prerequisites:

Analysis I and II; Measure and Integration Theory

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Pure Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective (MScData24)

Insurance Mathematics

Stefan Tappe in English

Lecture: Thu, 14–16 h, SR 404, Ernst-Zermelo-Str. 1

Content:

Insurance Mathematics has become an indispensable tool for insurance companies. It deals with the mathematical modeling of insured risks, the calculation of fair premiums for taking over such risks, and the computation of the required reserves.

In this lecture, we will start with life insurance mathematics, where we analyze several types of life insurance polices. Afterwards, we will proceed with risk theory, also known as non-life insurance mathematics. Here we consider portfolios of insured risks, where the number of claims and the claim sizes may be random.

Literature:

K.D. Schmidt: *Versicherungsmathematik*, Springer, 2006

Prerequisites:

Stochastik I, Probability Theory

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective in Data (MScData24)

Interactive Theorem Proving

Peter Pfaffelhuber, Assistant: Samuel Ayomide Adeosun in English
Lecture: Mon, 16–18 h, HS II, Albertstr. 23b
Exercise session: Wed, 16–18 h, HS II, Albertstr. 23b

Content:

In recent years, the Lean interactive theorem prover has gained a lot of interest among mathematicians. It comes as a programming language with a proof-interface, such that mathematical objects can be defined within the programming language, and the computer is able to check proofs it is given by the user in an interactive manner. This procedure is also called formalization of mathematics.

This course is based on my belief that such software will shape mathematical research in the future. It comes with a few aims:

* Introduce the Lean interactive theorem prover, including its huge mathematical library, mathlib. (This will be a hands-on part of the course.)
Provide some theoretical basics on theorem provers (above all, dependent types, functional programming, monads).
Some useful abstractions when interacting with theorem provers (e.g., order theory, filters, typeclasses).

Within the exercises, Lean code will be produced. Students will be asked to start a small formalization project by themselves.

Prerequisites:

Analysis, Linear Algebra

Remarks:

The course is offered in two versions. The above described material gives 6 ECTS. Together with a project, i.e. some self-chosen mathematical content formalized by the student, it gives 9 ECTS.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective in Data (MScData24)

Measure Theory

Peter Pfaffelhuber, Assistant: Samuel Ayomide Adeosun in English
Lecture: asynchronous (videos)
Tutorial: 2 hours, date to be determined and announced in class

Content:

Measure Theory is the foundation of advanced probability theory. In this course, we build on knowledge in analysis and provide all necessary results for later classes in statistics, probabilistic machine learning and stochastic processes. It contains set systems, constructions of measures using outer measures, the integral, and product measures.

Literature:

- H. Bauer: *Measure and Integration Theory*, de Gruyter, 2001.
- V. Bogatchev: *Measure Theory*, Springer, 2007.
- O. Kallenberg: *Foundations of Modern Probability Theory*, Springer, 2021.

Prerequisites:

Basic courses in analysis, and an understanding of mathematical proofs.

Remarks:

This is a self-study course with accompanying exercises that are marked.

Usable in the following modules:

Elective in Data (MScData24)

Numerical Optimal Control

Moritz Diehl in English
Tutorial / flipped classroom: Tue, 14–16 h, HS II, Albertstr. 23b
Lecture: asynchronous (videos)
Sit-in exam: date to be announced

Content:

The aim of the course is to give an introduction to numerical methods for the solution of optimal control problems in science and engineering. The focus is on both discrete time and continuous time optimal control in continuous state spaces. It is intended for a mixed audience of students from mathematics, engineering and computer science.

The course covers the following topics:

- Introduction to Dynamic Systems and Optimization
- Rehearsal of Newton-type methods and Numerical Optimization
- Algorithmic Differentiation
- Discrete Time Optimal Control
- Dynamic Programming
- Continuous Time Optimal Control
- Numerical Simulation Methods
- Hamilton–Jacobi–Bellmann Equation
- Pontryagin and the Indirect Approach
- Direct Optimal Control
- Real-Time Optimization for Model Predictive Control

The lecture is accompanied by intensive weekly programming exercises offered both in MATLAB and Python (6 ECTS) and an optional project (3 ECTS). The project consists in the formulation and implementation of a self-chosen optimal control problem and numerical solution method, resulting in documented computer code, a project report, and a public presentation.

Literature:

- M. Diehl, S. Gros: *Numerical Optimal Control*, lecture notes.
- J.B. Rawlings, D.Q. Mayne, M. Diehl: *Model Predictive Control*, 2nd Edition, Nobhill Publishing, 2017.
- J. Betts: *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming*, SIAM, 2010.

Prerequisites:

Required: Analysis I and II, Linear Algebra I and II
Recommended: Numerics I, Ordinary Differential Equations, Numerical Optimization

Remarks:

Together with the optional programming project, the course counts as a 9 ECTS lecture.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)

Theory and Numerics for Partial Differential Equations – Selected Nonlinear Problems

Sören Bartels, Luciano Sciaraffia in English
Lecture: Wed, 12–14 h, SR 226, Hermann-Herder-Str. 10
Tutorial: 2 hours, date to be determined and announced in class

Content:

The lecture addresses the development and analysis of numerical methods for the approximation of certain nonlinear partial differential equations. The considered model problems include harmonic maps into spheres and total-variation regularized minimization problems. For each of the problems, a suitable finite element discretization is devised, its convergence is analyzed and iterative solution procedures are developed. The lecture is complemented by theoretical and practical lab tutorials in which the results are deepened and experimentally tested.

Literature:

- S. Bartels: *Numerical methods for nonlinear partial differential equations*, Springer, 2015.
- M. Dobrowolski: *Angewandte Funktionalanalysis*, Springer, 2010.
- L.C. Evans: *Partial Differential Equations* (2nd edition), 2010.

Prerequisites:

'Introduction to Theory and Numerics for PDEs' or 'Introduction to PDEs'

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Applied Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective in Data (MScData24)

Topology of Fiber Bundles

Maximilian Stegemeyer in English
Lecture: Tue, 10–12 h, SR 404, Ernst-Zermelo-Str. 1
Tutorial: 2 hours, date to be determined and announced in class

Content:

Fiber bundles are an important concept in topology and geometry as they show up naturally in many situations. Fiber bundles are certain maps between spaces that behave "nice" locally. Global properties can therefore be understood by patching together local constructions. Next to their omnipresence in topology and geometry, the importance of fiber bundles is due to the fact that isomorphism classes of fiber bundles with a given fiber are a homotopy invariant of the underlying base space. Understanding these isomorphism classes is therefore a key task of algebraic topology. In this course we will introduce fiber bundles with a particular focus on vector bundles and principal fiber bundles. The main goal of the course is the classification of vector bundles. We shall further see some basics of K-theory and introduce characteristic classes of vector bundles. On the way we shall also get to know some concepts of homotopy theory.

Literature:

Dale Husemoller, *Fibre Bundles*, Springer, 2012

Norman Steenrod, *Topology of Fibre bundles*, Princeton University Press, 1951

Prerequisites:

Algebraic Topology
Basic knowledge in differential geometry is helpful, but not necessary.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Pure Mathematics (MSc14)
Mathematics (MSc14)
Specialisation Module (MSc14)
Elective (MSc14)
Elective (MScData24)

2a. Mathematics Education

Mathematics Education – Functions and Analysis

Katharina Böcherer-Linder in German
Thu, 9–12 h, SR 404, Ernst-Zermelo-Str. 1

Content:

Exemplary implementations of the theoretical concepts of central mathematical thought processes such as concept formation, modeling, problem solving and reasoning for the content areas of functions and analysis.
Barriers to understanding, pre-concepts, basic ideas, specific difficulties for the content areas of functions and analysis. Fundamental possibilities and limitations of media, in particular of computer-aided mathematical tools and their application for the content areas of functions and analysis. Analysis of individual mathematical learning processes and errors as well as development individual support measures for the content areas of functions and analysis.

Literature:

- R. Dankwerts, D. Vogel: *Analysis verständlich unterrichten*. Heidelberg: Spektrum, 2006.
- G. Greefrath, R. Oldenburg, H.-S. Siller, V. Ulm, H.-G. Weigand: *Didaktik der Analysis. Aspekte und Grundvorstellungen zentraler Begriffe*. Berlin, Heidelberg: Springer 2016.

Prerequisites:

Introduction to Mathematics Education
Knowledge about analysis and numerics

Remarks:

The two parts can be completed in different semesters, but have a joint final exam, which is offered every semester and written after completing both parts.

Usable in the following modules:

Mathematics Education for Specific Areas of Mathematics (MEd18, MEH21, MEB21)

Mathematics Education – Probability Theory and Algebra

Anika Dreher in German
Fri, 9–12 h, SR 404, Ernst-Zermelo-Str. 1

Content:

Exemplary implementations of the theoretical concepts of central mathematical thought processes such as concept formation, modeling, problem solving and reasoning for the content areas of stochastics and algebra.
Barriers to understanding, pre-concepts, basic ideas, specific difficulties for the content areas of stochastics and algebra.
Basic possibilities and limitations of media, especially computer-based mathematical tools and their application for the content areas of stochastics and algebra.
Analysis of individual mathematical learning processes and errors as well as development individual support measures for the content areas of stochastics and algebra.

Literature:

- G. Malle: *Didaktische Probleme der elementaren Algebra*. Braunschweig, Wiesbaden: Vieweg 1993.
- A. Eichler, M. Vogel: *Leitidee Daten und Zufall. Von konkreten Beispielen zur Didaktik der Stochastik*. Wiesbaden: Vieweg 2009.

Prerequisites:

Introduction to Mathematics Education
Knowledge from stochastics and algebra

Remarks:

The two parts can be completed in different semesters, but have a joint final exam, which is offered every semester and written after completing both parts.

Usable in the following modules:

Mathematics Education for Specific Areas of Mathematics (MEd18, MEH21, MEB21)

Introduction to Mathematics Education

Katharina Böcherer-Linder in German
Mon, 10–12 h, SR 226, Hermann-Herder-Str. 10
Tutorial: 2 hours, various dates
Sit-in exam: date to be announced

Content:

Mathematics didactic principles and their learning theory foundations and possibilities of teaching implementation (also e.g. with the help of digital media).
Theoretical concepts on central mathematical thinking activities such as concept formation, modeling, problem solving and reasoning.
Mathematics didactic constructs: Barriers to understanding, pre-concepts, basic ideas, specific difficulties with selected mathematical content.
Concepts for dealing with heterogeneity, taking into account subject-specific characteristics particularities (e.g. dyscalculia or mathematical giftedness).
Levels of conceptual rigour and formalization as well as their age-appropriate implementation.

Prerequisites:

Required: Basics lectures (Analysis, Linear Algebra)

The course 'Introduction to Mathematics Education' is therefore recommended from the 4th semester at the earliest.

Remarks:

The course is compulsory in the teacher education track of the Polyvalent Dual Major Bachelor's degree programme. It is made up of lecture components and parts with exercise and student seminar character. The three forms of teaching cannot be not be completely separated from each other. Attendance at the "Didactic Seminar" (approximately fortnightly, tuesday evenings, 19:30) is expected!
This course is only offered in German.

Usable in the following modules:

(Introduction to) Mathematics Education (2HfB21, MEH21, MEB21)
Introduction to Mathematics Education (MEdual24)

Mathematics education seminar: Media Use in Teaching Mathematics

Jürgen Kury — in German

Seminar: Wed, 15–18 h, SR 404, Ernst-Zermelo-Str. 1

Content:

The use of teaching resources in mathematics lessons is becoming increasingly important, both in terms of lesson planning and lesson delivery. Against the backdrop of constructivist learning theories, it is evident that the thoughtful use of modular mathematics systems in the classroom enhances the long-term effectiveness of teaching. Particular emphasis is placed on giving greater weight to deep structures. These include cognitive activation, constructive support for learning and the consolidation of learning processes. Digital media can be used specifically here to deepen mathematical understanding, stimulate thinking processes and promote sustainable learning.

Artificial intelligence (AI) is increasingly being incorporated into mathematics teaching. This student seminar explores how AI-based systems – such as automated diagnostic tools, adaptive learning aids and intelligent tutoring systems – can support teachers. The aim is to reflect on the opportunities and challenges of using AI, and to test and evaluate specific ways in which it can be used in the classroom.

The student seminar aims to equip students with the necessary decision-making and practical skills to prepare future mathematics teachers for their professional careers. Starting with initial considerations regarding lesson planning, computers and tablets are then examined in terms of their respective educational potential and tested during a classroom visit with pupils.

Students are expected to develop lesson sequences, which are then tested with pupils and reflected upon.

Prerequisites:

Recommended: Basic courses in mathematics

GeoGebra Account (can be created in the student seminar)

Usable in the following modules:

Supplementary Module in Mathematics Education (MEd18, MEH21, MEB21)

Mathematics education seminars at Freiburg University of Education

Lecturers of the University of Education Freiburg in German

Remarks:

For the module "Fachdidaktische Entwicklung", suitable courses can also be completed at the PH Freiburg, if places are available there. To find out whether courses are suitable, please discuss in advance with Ms. Böcherer-Linder; you must check whether places are available if you are interested in a course from the lecturers if you are interested in a course.
Most suitable courses will be offered in German.

Usable in the following modules:

Supplementary Module in Mathematics Education (MEd18, MEH21, MEB21)

Module "Research in Mathematics Education"

Lecturers of the University of Education Freiburg, Anselm Strohmaier in German

Part 1: Seminar 'Development Research in Mathematics Education – Selected Topics': Mon, 14–16 h, 301, KG 4, University of Education Freiburg,
Please refer to the PH Freiburg course catalogue for any last-minute time or room changes.
Part 2: Seminar 'Research Methods in Mathematics Education': Mon, 10–13 h, 010, Pavillon 3, University of Education Freiburg,
Starting on 22 December 2025. Please refer to the PH Freiburg course catalogue for any last-minute time or room changes.
Part 3: Master's thesis seminar: Development and Optimisation of a Research Project in Mathematics Education
Appointments by arrangement

Content:

The three related courses of the module prepare students for an empirical Master thesis in mathematics didactics. The course is jointly designed by all professors at the PH with mathematics didactics research projects at secondary levels 1 and 2 and is carried out by one of these researchers. Afterwards, students have the opportunity to start a Master thesis with one of these supervisors - usually integrated into larger ongoing research projects.

The first course of the module provides an introduction to strategies of empirical didactic research (research questions, research status, research designs). Students deepen their skills in scientific research and the evaluation of subject-specific didactic research. In the second course (in the last third of the semester) students are introduced to central qualitative and quantitative research methods through concrete work with existing data (interviews, student products, experimental data), students are introduced to central qualitative and quantitative research methods. The third course is an accompanying student seminar for the Master thesis.

The main objectives of the module are the ability to receive mathematics didactic research in order to clarify questions of practical relevance and to plan an empirical mathematics didactics Master thesis. It will be held as a mixture of seminar, development of research topics in groups and active work with research data. Recommended literature will be depending on the research topics offered within the respective courses. The parts can also be attended in different semesters, for example part 1 in the second Master semester and part 2 in the compact phase of the third Master semester after the practical semester.

Remarks:

Three-part module for M.Ed. students who would like to write a didactic Master thesis in mathematics. Participation only after personal registration by the end of the lecture period of the previous semester in the Department of Didactics. Admission capacity is limited.

Pre-registration: If you would like to take part in this module, please register by 30 September 2026 by e-mail to didaktik@math.uni-freiburg.de and to Anselm Strohmaier. This course will only be offered in German.

Usable in the following modules:

Research in Mathematics Education (MEd18, MEH21, MEB21)

2b. Tutorial Module

Learning by Teaching

in German

Raum 232, Ernst-Zermelo-Str. 1, ,

Content:

What characterizes a good tutorial? This question will be discussed in the first workshop and tips and suggestions will be given. Experiences will be shared in the second workshop.

Remarks:

Prerequisite for participation is a tutoring position for a lecture of the Mathematical Institute in the current semester (at least one two-hour or two one-hour tutorial groups over the whole semester).
Can be used twice in the M.Sc. programme in Mathematics.
This course is only offered in German.

This course is worth 3 credits as an elective module.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Elective (BSc21)
Elective (MSc14)
Elective (MScData24)
Supplementary Module in Mathematics (MEd18)

2c. Programming Exercises

Programming exercises for 'Introduction to Theory and Numerics of Partial Differential Equations'

Alen Kushova, Michael Růžička in English
Programming exercise: 2 hours, date to be determined

Content:

The objective of the programming exercises is the development of a small finite element software library in a programming language of the student's choice (e.g. MATLAB, Python, Julia, C/C++). The sessions are organized as a sequence of key implementation including numerical integration on simplicial meshes, assembly of mass and stiffness matrices, treatment of boundary conditions and error estimation.

Students who plan to write an admission thesis, master's thesis, or diploma thesis in Applied Mathematics are encouraged to participate in the practical exercises. Basic programming experience (e.g. MATLAB, Python, Julia, C/C++) is required.

Prerequisites:

See the lecture – additionally: programming knowledge.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Elective (BSc21)
Supplementary Module in Mathematics (MEd18)
Elective (MSc14)
Elective (MScData24)

Programming exercises for 'Numerics'

Sören Bartels, Assistant: Jonathan Brugger in German
Programming exercise: 2 hours, various dates

Content:

In the programming exercise accompanying the Numerics I (first term) lecture the algorithms developed and analyzed in the lecture are put into practice and and tested experimentally. The implementation is carried out in the programming languages Matlab, C++ and Python. Elementary programming knowledge is assumed.

Prerequisites:

See the lecture *Numerics I* (which should be attended in parallel or should already have been completed).
Additionally: Elementary programming knowledge.

Usable in the following modules:

Computer Exercise (2HfB21, MEH21, MEB21)
Elective (Option Area) (2HfB21)
Numerics (BSc21)
Supplementary Module in Mathematics (MEd18)

Programming exercises for 'Theory and Numerics of Partial Differential Equations – Selected Nonlinear Problems'

Sören Bartels, Assistant: Luciano Sciaraffia in English
Programming exercise: 2 hours, date to be determined

Content:

In the programming exercises accompanying the lecture 'Theory and Numerics for Partial Differential Equations – Selected Nonlinear Problems', the algorithms developed and analyzed in the lecture are implemented and tested experimentally. The implementation can be carried out in the programming languages Matlab, C++ or Python. Elementary programming knowledge is assumed.

Literature:

S. Bartels: *Numerical methods for nonlinear partial differential equations*, Springer, 2015.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Elective (BSc21)
Supplementary Module in Mathematics (MEd18)
Elective (MSc14)
Elective (MScData24)

Python for Data Analysis

Sebastian Stroppel in English

Content:

This course is designed for students without prior knowledge in programming, but students who have already taken a first programming course might benefit as well. We will start with basic syntax and the standard library of python, including data types, functions, loops, regular expressions, and interacting with the operating system. For data analysis we learn dataframes using packages such as pandas (and relatives), see how we can interact with freely available APIs, make plots using matplotlob, and use numpy and scipy for standard procedures including numerical computations.

Within this course, you will pick a programming task of your interest, and implement your ideas based on your gained knowledge.

Prerequisites:

none

Remarks:

Cannot be credited together with the Programming Exercise in Stochastics in Python.

Usable in the following modules:

Elective (MScData24)
Computer Exercise (2HfB21, MEH21, MEB21)

3a. Introductory student seminars

Introductory student seminar: A Selection of Famous Mathematicians

Annette Huber-Klawitter, Assistant: Riccardo Tosi in German

Remarks:

Remaining places on the M.Ed. student seminar may be allocated as introductory student seminar; for further details, see 'Student seminars', in particular the pre-registration procedures and the preliminary meeting date!

Usable in the following modules:

Introductory Student Seminar (2HfB21, BSc21, MEH21, MEB21)

Introductory student seminar: Ordinary Differential Equations

Diyora Salimova, Assistant: Ilkhom Mukhammadiev Talk/participation possible in German and English
Seminar: Tue, 12–14 h, SR 226, Hermann-Herder-Str. 10
Preregistration: by 19 July 2026 by email to Diyora Salimova
Preliminary seminar meeting 22.07., 11:00–12:00, SR 226, Hermann-Herder-Str. 10, ,

Content:

In this introductory student seminar we will explore several aspects of Ordinary Differential Equations (ODEs), a fundamental area of mathematics with widespread applications across natural sciences, engineering, economics, and beyond. Students will engage actively by presenting and discussing various topics, including existence and uniqueness theorems, stability analysis, linear systems, nonlinear dynamics, and numerical methods for solving ODEs. Participants will enhance their analytical skills and deepen their theoretical understanding by studying classical problems and contemporary research directions.

Literature:

- J. W. Prüss, M. Wilke: *Gewöhnliche Differentialgleichungen und dynamische Systeme*, Grundstudium Mathematik.
- R. D'Ambrosio: *Numerical Approximation of Ordinary Differential Problems*
- M. Hermann und M. Saravi: *Eine Einführung in gewöhnliche Differentialgleichungen*

Prerequisites:

Analysis I and II, Linear Algebra I and II

Usable in the following modules:

Introductory Student Seminar (2HfB21, BSc21, MEH21, MEB21)

Introductory student seminar: Hierarchies of Formal Grammars

Heike Mildenberger in German

Seminar: Tue, 16–18 h, SR 125, Ernst-Zermelo-Str. 1

Preliminary seminar meeting 20.07., 13:00–14:00, Room 313, Ernst-Zermelo-Str. 1, , No prior registration necessary. If you are unable to attend yourself, you are welcome to send a representative.

Content:

In the 1930s, Turing, Church, Kleene and others clarified what we now understand as a computable set of words (also known as a computable language). Computable sets are Turing-computable sets and are also referred to as recursive sets. There is also a simple description in terms of number theory. In 1956, Noam Chomsky published a seminal paper on computability theory, in which he introduced more precise concepts of computability. These are known as regular, context-free and context-sensitive grammars. The general Chomsky grammar coincides with the class of recursively enumerable sets.

Later, correspondences were found between certain computational powers of automata and the levels of grammar. Following theorems by Myhill and Nerode in 1957 and Rabin and Scott in 1958, the paper [1] by Bar-Hillel, Perles and Shamir, containing the pumping lemma from 1961, is regarded as a further important milestone.

In this introductory student seminar, we will study the relationships between the writing power of a Turing machine and the various automata on the one hand, and the grammar hierarchy on the other. Those who are interested may consult the original literature. We will mainly be working with the Springer textbook by Erk and Priese [2], which is available online via the University Library and in which all correspondences are shown. The classic textbook by Hopcroft, Motwani and Ullman, *Introduction to Automata Theory, Formal Languages and Complexity Theory* [3], is also very suitable. However, context-sensitive grammars (which constitute L_1 in the Chomsky hierarchy of L_3, L_2, L_1, L_0 – where a lower index denotes a larger class) are scarcely covered in that text.

Literature:

1 Yehoshua Bar-Hillel, Michael Perles, and Eli Shamir. *On formal properties of simple phrase structure grammars.* Z. Phonetik Sprachwiss. Kommunikat., 14:143–172, 1961.

2 Katrin Erk and Lutz Priese. *Theoretische Informatik. Eine umfassende Einführung.* Springer-Lehrb. Berlin: Springer, 4., erweit. Aufl. edition, 2017.

3 John E. Hopcroft, Rajeev Motwani, and Jeffrey D. Ullman. *Einführung in die Automatentheorie, formale Sprachen und Komplexitätstheorie.* Pearson Studium, 2., überarbeitete Aufl. edition, 2002.

Prerequisites:

Proofs by induction

Usable in the following modules:

Introductory Student Seminar (2HfB21, BSc21, MEH21, MEB21)

Introductory student seminar: Mathematics in Everyday Life

Sebastian Goette, Assistant: Mikhail Tëmkin in German
Seminar: Wed, 10–12 h, SR 125, Ernst-Zermelo-Str. 1
Preliminary seminar meeting 16.07., 13:00–14:00, SR 125, Ernst-Zermelo-Str. 1, , no prior registration required

Content:

In everyday life, mathematics helps us to describe, understand and, often, solve problems in a wide variety of fields. Examples include cryptography (e.g. RSA codes), coding (e.g. audio digitalisation or QR codes), and technical devices (e.g. calculators, navigation systems). Mathematics also plays a role in the social sciences, for example in game theory within economics.

In this introductory student seminar, we aim, on the one hand, to familiarise ourselves with these applications and their connection to mathematics. On the other hand, we aim to describe the problems as accurately as possible using mathematical methods and, ideally, to solve them as well.

The recommended reading list is intended only as a starting point; participants are expected to find further sources themselves.

Literature:

This is specified in the syllabus and is intended only as an initial guide; further literature research is encouraged.

Prerequisites:

Analysis I, II, Linear Algebra I, II. Further prior knowledge may be required for individual lectures; this is indicated in the programme.

Remarks:

Participants are welcome to suggest their own topics, provided they fit within the scope of the introductory student seminar. In this case, please contact S. Goette in good time before the preliminary meeting.

Usable in the following modules:

Introductory Student Seminar (2HfB21, BSc21, MEH21, MEB21)

3b. Student seminars

M.Ed.-student seminar (after practical teaching semester): A Selection of Famous Mathematicians

Annette Huber-Klawitter, Assistant: Riccardo Tosi in German

Block seminar after the school practical semester The seminar will take place during the week of 15–19 February 2027.

Preregistration: List available from Ms Frei, Room 421, Ludmilla Frei

Preliminary seminar meeting 13.07., 12:30–13:30, SR 404, Ernst-Zermelo-Str. 1, ,

Content:

Each lecture in this student seminar will introduce a mathematician whose results have proved fundamental to the development of mathematics. The aim of this course is therefore not only to explain important theorems that have shaped the course of this discipline, but also to get to know the people behind the proofs. Following an overview of the life of the mathematician in question and, where relevant, the historical context, one of their key results will be explained, together with the corresponding proof. Examples of the topics covered in the student seminar include analysis (Fourier), algebra (Noether), mathematical physics (Kowalewskaja), set theory (Cantor, Zermelo) and number theory (Lindemann, Germain).

Prerequisites:

A basic knowledge of Linear Algebra and Analysis.

Remarks:

The student seminar is primarily intended for M.Ed. students. Any remaining places may be allocated to introductory student seminar students.

It will be held as a block seminar in January/February 2027.

Usable in the following modules:

Introductory Student Seminar (2HfB21, BSc21, MEH21, MEB21)
Supplementary Module in Mathematics (MEd18)

Student Seminar: Representation Theory

Wolfgang Soergel, Assistant: Xier Ren Talk/participation possible in German and English
Seminar: Thu, 10–12 h, SR 125, Ernst-Zermelo-Str. 1
Preregistration: by e-mail to Wolfgang Soergel
Preliminary seminar meeting 15.07., 15:15, SR 414, Ernst-Zermelo-Str. 1, ,

Content:

This student seminar will focus on irreducible representations of non-compact Lie groups and, in particular, on the transition to the so-called Harish-Chandra modules and the significance of the category $\mathcal{O}$. The student seminar builds on the lecture on Lie groups and the student seminar on semi-simple Lie algebras from the summer term.

Prerequisites:

Lecture on Lie groups or the student seminar on semi-simple Lie algebra.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Mathematical Seminar (BSc21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Mathematical Seminar (MSc14)
Elective (MSc14)
Elective (MScData24)

Student Seminar: Eigenvalue Problems

Guofang Wang, Assistant: Xuwen Zhang Talk/participation possible in German and English
Seminar: Wed, 16–18 h, SR 125, Ernst-Zermelo-Str. 1
Preliminary seminar meeting 22.07., 16:15, SR 125, Ernst-Zermelo-Str. 1, , no prior registration necessary!

Content:

Eigenvalues play an important role to understand an operator. In this student seminar, we learn basic knowledge about eigenvalues of the Laplace operator with certain boundary condition and compute its eigenvalues and also eigenfunctions for special domains.

Literature:

Chavel, Eigenvalues in Riemannian geometry

Prerequisites:

Analysis III, Functional analysis

Usable in the following modules:

Elective (Option Area) (2HfB21)
Mathematical Seminar (BSc21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Mathematical Seminar (MSc14)
Elective (MSc14)
Elective (MScData24)

Student seminar: Medical Data Science

Harald Binder Talk/participation possible in German and English
Seminar: Wed, 10:15–11:30 h, HS Medizinische Biometrie, 1. OG, Stefan-Meier-Str. 26
Preregistration: by email to Olga Sieber
Preliminary seminar meeting 22.07., 13:30–15:30, HS Medizinische Biometrie, 1. OG, Stefan-Meier-Str. 26, ,

Content:

To answer complex biomedical questions from large amounts of data, a wide range of analysis tools is often necessary, e.g. deep learning or general machine learning techniques, which is often summarized under the term "Medical Data Science". Statistical approaches play an important role as the basis for this. A selection of approaches is to be presented in the student seminar lectures that are based on recent original work. The exact thematic orientation is still to be determined.

Literature:

Information on introductory literature is given in the preliminary meeting.

Prerequisites:

Good knowledge of probability theory and mathematical statistics.

Remarks:

The student seminar can serve as basis for a bachelor's or master's thesis.

Usable in the following modules:

Elective (Option Area) (2HfB21)
Mathematical Seminar (BSc21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Mathematical Seminar (MSc14)
Elective (MSc14)
Mathematical Seminar (MScData24)
Elective in Data (MScData24)

Student seminar: Optimal Transport

Patrick Dondl, Assistant: Oliver Suchan in English

Seminar: Wed, 14–16 h, SR 226, Hermann-Herder-Str. 10

Preliminary seminar meeting 16.07., 12:00, Bibliotheksraum 216, Hermann-Herder-Str. 10, , no prior registration necessary!

Content:

Optimal transport concerns the problem, going back to Monge (1781), of moving one distribution of mass onto another at minimal cost. Kantorovich's relaxation recasts this as a linear problem with a rich duality theory and equips the space of probability measures with the Wasserstein distances. In this student seminar we develop the theory from the Monge–Kantorovich problem and Kantorovich duality through Brenier's theorem on the existence and structure of optimal maps and its link to the Monge–Ampère equation, geodesics and displacement convexity in Wasserstein space, up to the connection with gradient flows (Otto calculus, JKO scheme). Time permitting, we also treat the computational side (entropic regularization, Sinkhorn's algorithm). Optimal transport has become a central tool in the analysis of nonlinear PDE, in geometry and probability, and increasingly in imaging and data science.

Literature:

- C. Villani, *Topics in Optimal Transportation*, Graduate Studies in Mathematics 58, AMS, 2003.
- F. Santambrogio, *Optimal Transport for Applied Mathematicians, Progress in Nonlinear Differential Equations and Their Applications* 87, Birkhäuser, 2015.
- G. Peyré, M. Cuturi, *Computational Optimal Transport, Foundations and Trends in Machine Learning* 11(5–6), 2019.

Prerequisites:

Functional Analysis

Usable in the following modules:

Elective (Option Area) (2HfB21)
Mathematical Seminar (BSc21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Mathematical Seminar (MSc14)
Elective (MSc14)
Mathematical Seminar (MScData24)
Elective in Data (MScData24)

Student seminar: Orlicz Spaces

Michael Růžička Talk/participation possible in German and English
Seminar: Mon, 14–16 h, SR 127, Ernst-Zermelo-Str. 1
Preliminary seminar meeting 20.07., 13:00–14:00, SR 127, Ernst-Zermelo-Str. 1, , no prior regristration necessary!

Content:

In this student seminar, we will generalise the theory of Lebesgue spaces $L^p(\Omega)$ to Orlicz spaces $L^\phi(\Omega)$. These spaces play a major role in the theory and numerical analysis of nonlinear partial differential equations.

Literature:

L. Pick, A. Kufner, O. John, S. Fučik: *Function spaces* Vol.1, De Gruyter 2013

Prerequisites:

Analysis I–III, and some Functional Analysis

Usable in the following modules:

Elective (Option Area) (2HfB21)
Mathematical Seminar (BSc21)
Compulsory Elective in Mathematics (BSc21)
Supplementary Module in Mathematics (MEd18)
Mathematical Seminar (MSc14)
Elective (MSc14)
Mathematical Seminar (MScData24)
Elective in Data (MScData24)

Student seminar: Data-Driven Medicine from Routine Data

Nadine Binder Talk/participation possible in German and English
Seminar: Thu, 16:30–18 h, HS Medizinische Biometrie, 1. OG, Stefan-Meier-Str. 26,
or by appointment
Preregistration: by email to PD Dr. Nadine Binder
Preliminary seminar meeting 23.07., 13:00, HS Medizinische Biometrie, 1. OG, Stefan-Meier-Str. 26, ,
Preliminary seminar meeting 06.10., 16:30, HS Medizinische Biometrie, 1. OG, Stefan-Meier-Str. 26, ,

Content:

The student seminar is a journal-club style meeting where we critically read and discuss recent papers that use routine health-care data. You'll dissect the statistical or computational models, incl. survival analysis, causal-effects methods, or machine learning, that turn raw diagnoses, labs, or medication records into clinical insights. You'll work individually or potentially in pairs with medical students to prepare a presentation that summarizes the study, evaluates its methodology, and reflects on how the mathematics could be refined or applied elsewhere. The student seminar format will allow you to sharpen your ability for interpreting quantitative research, bridge theory with practice, and experience the interdisciplinary dialogue that drives modern evidence-based medicine.

Literature:

References will be provided during the course.

Prerequisites:

None that go beyond admission to the degree programme.

Usable in the following modules:

Mathematical Seminar (MScData24)
Elective in Data (MScData24)

Graduate Student Speaker Series

in English

Seminar: Mon, 14–16 h, SR 226, Hermann-Herder-Str. 10

Content:

In the Graduate Student Speaker Series, students of the M.Sc. degreee programme 'Mathematics in Data and Technology' talk about their Master's thesis or their programming projects, and the lecturers of the programme talk about their fields of work.

Usable in the following modules:

Graduate Student Speaker Series (MScData24)